Martin Angell

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Education

New York University, Courant Institute — New York, NY
Bachelor of Arts in Computer Science and Mathematics — *May 2027*

- **Cumulative GPA:** 3.85/4.00
- **CS Coursework:** Computer Graphics, Computer Simulation, Numerical Computing, Theory of Computation, Programming Languages, Operating Systems
- **Math Coursework:** Probability Theory, ODEs, Real Analysis, Abstract Algebra

Research Experience

Computer Graphics Research — New York University
Undergraduate Researcher, advised by Prof. Gizem Kayar — *Fall 2024 – Present*

- Solved the core problem of generating matched low-resolution/high-resolution Smoothed Particle Hydrodynamics (SPH) simulation pairs for training neural fluid-upscaling models.
- Built in-house fluid simulators and renderers using Divergence-Free Smoothed Particle Hydrodynamics and screen-space rendering. This included implementing eight classical SPH upsampling papers.
- Developed online tools used to train future fluid simulation researchers for our lab.

Publications

[1] **Martin Angell***, Matias Ortiz*, and Gizem Kayar. "FluidScale: A Machine Learning Model for Upscaling Screen Space Fluids for Games." *In preparation for SIGGRAPH 2027.* (*equal contribution)

[2] **Martin Angell** and Gizem Kayar. "LearnSPH: A Tool for Teaching the Basics of Smoothed Particle Hydrodynamics in Graphics Research." *SIGCSE TS 2026*, St. Louis, MO. Poster, presented in person. doi:10.1145/3770761.3777283

Teaching

Grader, Computer Graphics — New York University
Department of Computer Science — *Fall 2025, Spring 2026, Fall 2026*

- Evaluated five to six Blender-based graphics projects for 20-35 students per semester.
- Created an assignment evaluating students' understanding of basic fluid simulation and rigid bodies.
- Handled regrade requests.

Grader, Computer Systems Organization — New York University
Department of Computer Science — *Fall 2024, Spring 2025*

- Evaluated six written and programming assignments and six quizzes for 50 students per semester.
- Worked alongside another grader to ensure consistent and fair grading policies.

Honors and Grants

Dean's Undergraduate Research Fund, New York University — *2025*
Awarded for the development of LearnSPH

Dean's List, College of Arts and Science, New York University — *2024–2025, 2025–2026*

Skills

Programming Languages: C++, Java, Python, JavaScript
Graphics & Simulation: OpenGL, CUDA, Blender, GLSL
Tools: Git, LaTeX, NodeJS
Languages: Conversational French